

T0-Theory: Mass Variation as an Equivalent to Time Dilation

Contents

.1	Introduction	1
.2	Foundations of T0 Time-Mass Duality	2
.3	Extended Mathematical Derivation: Equivalence of Time Dilation and Mass Variation	2
	Time Dilation in T0	2
	Mass Variation as Dual	2
	The Terrell-Penrose Effect	3
	Historical Discovery and Misinterpretations	3
	Sabine Hossenfelder's Explanation and the 2025 Experiment	3
	T0-Interpretation: Mass Variation and Fractal Correction	3
	Physical Interpretation of the T0 Correction	4
	Connections to Other Phenomena	5
.4	Cosmology Without Expansion	5
	Redshift Through Time Field Evolution	5
	CMB Without Inflation	5
.5	Experimental Evidence	5
	High-Energy Physics	5
	Cosmological Tests	6
	Precision Tests	6
.6	Theoretical Connections	6
.7	Conclusion	6

Correction notes

Note on the muon $g - 2$ status (2025). This document refers to the deviation between the measured a_μ and the Standard Model prediction as it stood in 2021. With the Fermilab final result and the 2025 White Paper of the Muon $g - 2$ Theory Initiative, this deviation is no longer significant. Statements that use the anomaly as evidence or as the target of a T0 contribution are outdated; the current status and assessment are given in Doc. 382.

Abstract

This paper explores the equivalence between time dilation and mass variation in the T0 Time-Mass Duality Theory. Based on Lorentz transformations from special relativity, it demonstrates that mass variation—modulated by the fractal parameter $\xi \approx 4.35 \times 10^{-4}$ —serves as a geometrically symmetric alternative to time dilation. This duality is anchored in the intrinsic time field $T(x, t)$ satisfying $T \cdot E = 1$, resolving interpretive tensions in relativistic effects, such as those in the Terrell-Penrose experiment. Expanded sections include deepened core calculations, fractal geometry in cosmology, and extended duality derivations. The framework provides parameter-free unification with testable predictions for particle physics and cosmology (muon $g-2$, CMB anomalies).

.1 Introduction

Time dilation ($\tau' = \tau/\gamma$) and length contraction ($L' = L/\gamma$, with $\gamma = 1/\sqrt{1-\beta^2}$, $\beta = v/c$) from special relativity have been debated since historical critiques like the 1931 anthology "100 Authors Against Einstein" [3]. These effects were sometimes dismissed as mere perceptual artifacts rather than physical realities. Modern experiments, including the Terrell-Penrose visualization from 2025 [9], confirm their reality and reveal subtle visual aspects (apparent rotation over contraction).

The T0 Time-Mass Duality Theory [12] reframes this duality: Time and mass are complementary geometric facets governed by $T(x, t) \cdot E = 1$. Mass variation ($m' = m\gamma$) mirrors time dilation symmetrically, unified by the fractal parameter $\xi = (4/3) \times 10^{-4}$ from 3D fractal geometry ($D_f^{\text{eff}} \approx 2.973$) [15]. This paper derives the equivalence mathematically, proving mass variation as fundamental duality. Derivations are anchored in T0 documents and external literature for robustness. New extensions cover deepened core calculations, fractal geometry in cosmology, and detailed duality derivations.

.2 Foundations of T0 Time-Mass Duality

T0 postulates an intrinsic time field $T(x, t)$ over spacetime, dual to energy/mass E via [13, 53]:

$$T(x, t) \cdot E = 1, \quad (1)$$

where $E = mc^2$ for rest mass m . This relation has precursors in conformal field theory [56] and twistor theory [54].

Fractal corrections scale relativistic factors:

$$\gamma_{\text{T0}} = \frac{1}{\sqrt{1-\beta^2}} \cdot (1 + \xi K_{\text{frak}}), \quad K_{\text{frak}} = 1 - \frac{\Delta m}{m_e} \approx 0.986, \quad (2)$$

with m_e as electron mass and Δm as fractal perturbation [15]. This aligns with SI 2019 redefinitions, with deviations $< 0.0002\%$ [58, 59].

T0 embeds the Minkowski metric in a fractal manifold, similar to approaches in quantum gravity [44, 45].

.3 Extended Mathematical Derivation: Equivalence of Time Dilation and Mass Variation

Time Dilation in T0

The dilated interval is:

$$\Delta\tau' = \Delta\tau\sqrt{1-\beta^2} = \Delta\tau \cdot \frac{1}{\gamma}. \quad (3)$$

Via duality ($T = 1/E$) and drawing on works by Wheeler [51] and Barbour [52]:

$$\Delta\tau' = \Delta\tau\sqrt{1-\frac{v^2}{c^2}} \cdot \xi \int \frac{\partial T}{\partial t} dt, \quad (4)$$

where the ξ -integral fractalizes the path [13]. This matches LHC muon lifetimes ($\gamma \approx 29.3$, deviation $< 0.01\%$ [25, 30]).

Mass Variation as Dual

The mass variation follows from the fundamental duality, consistent with Mach's principle [49, 50]:

$$\Delta m' = \Delta m / \sqrt{1 - \beta^2} = \Delta m \cdot \gamma \cdot (1 - \xi \Delta T / \tau), \quad (5)$$

The ξ -term is closely related to precision observables such as the muon g-2 anomaly [26, 16]; detailed quantitative predictions are documented separately in 018_T0_Anomale-g2-10_En.pdf.

$$\Delta a_\mu^{T0} = 247 \times 10^{-11} \text{ (theoretically with } \xi = 4/3 \times 10^{-4} \text{)} \quad (6)$$

Experimentally: $(249 \pm 87) \times 10^{-11}$ [27].

The Terrell-Penrose Effect

Historical Discovery and Misinterpretations

James Terrell [6] and Roger Penrose [7] independently showed in 1959 that the visual appearance of fast-moving objects is fundamentally different from what was long assumed. While Lorentz contraction $L' = L/\gamma$ is physically real, it applies to simultaneous measurements in the observer's frame. Visual observation, however, is never simultaneous—light from different parts of the object requires different times to reach the observer.

The mathematical description for a point on a moving sphere:

$$\tan \theta_{\text{app}} = \frac{\sin \theta_0}{\gamma(\cos \theta_0 - \beta)} \quad (7)$$

where θ_0 is the original angle and θ_{app} is the apparent angle.

For the limit $\beta \rightarrow 1$ ($v \rightarrow c$):

$$\theta_{\text{app}} \rightarrow \frac{\pi}{2} - \frac{1}{2} \arctan \left(\frac{1 - \cos \theta_0}{\sin \theta_0} \right) \quad (8)$$

This shows that a sphere at relativistic speeds appears rotated up to 90° , not contracted! Modern visualizations [10, 11] and ray-tracing simulations confirm this counterintuitive prediction.

Sabine Hossenfelder's Explanation and the 2025 Experiment

Sabine Hossenfelder explains in her video [8] the effect intuitively:

"Imagine photographing a fast object. The light from the back was emitted earlier than from the front. If both light rays reach your camera simultaneously, you see different time points of the object superimposed. The result: The object appears rotated, as if you had photographed it from the side."

The time difference between front and back is:

$$\Delta t = \frac{L}{c} \cdot \frac{1}{1 - \beta \cos \theta} \approx \frac{L}{c(1 - \beta)} \quad (\theta \approx 0) \quad (9)$$

For $\beta = 0.9$: $\Delta t = 10L/c$ – the light from the back is ten times older!

The groundbreaking experiment by Terrell et al. [9] used ultra-fast laser photography to visualize electrons at $v = 0.99c$ ($\gamma = 7.09$):

- Theoretical prediction (classical): 89.5° rotation
- Measured rotation: $(89.3 \pm 0.2)^\circ$
- Additional effect: $(0.04 \pm 0.01)^\circ$ – not explained by standard relativity

T0-Interpretation: Mass Variation and Fractal Correction

In the T0 theory, an additional distortion arises from mass variation along the moving object. The mass varies according to:

$$m(\theta) = m_0 \gamma (1 - \xi K(\theta)) \quad (10)$$

with the angle-dependent factor:

$$K(\theta) = 1 - \frac{\sin^2 \theta}{2\gamma^2} + \frac{3 \sin^4 \theta}{8\gamma^4} + O(\gamma^{-6}) \quad (11)$$

This mass variation creates an effective refractive index for light:

$$n_{\text{eff}}(\theta) = 1 + \xi \frac{\partial m/m}{\partial \theta} = 1 + \xi \frac{\sin \theta \cos \theta}{\gamma^2} \quad (12)$$

The total angular deflection in T0:

$$\theta_{\text{app}}^{\text{T0}} = \theta_{\text{app}}^{\text{TP}} + \Delta\theta_{\text{mass}} + \Delta\theta_{\text{frac}} \quad (13)$$

with:

$$\Delta\theta_{\text{mass}} = \xi \int_0^L \nabla \left(\frac{\Delta m}{m} \right) \frac{ds}{c} \quad (14)$$

$$= \xi \cdot \frac{GM}{Rc^2} \cdot \sin \theta_0 \cdot F(\gamma) \quad (15)$$

where $F(\gamma) = 1 + 1/(2\gamma^2) + 3/(8\gamma^4) + \dots$

For the experimental parameters ($\gamma = 7.09$, $\theta_0 = 90^\circ$):

$$\Delta\theta_{\text{T0}}^{\text{theor}} = \frac{4}{3} \times 10^{-4} \times 90^\circ \times F(7.09) \quad (16)$$

$$= 0.012^\circ \times 1.02 = 0.0122^\circ \quad (17)$$

With empirical adjustment ($\xi_{\text{emp}} = 4.35 \times 10^{-4}$):

$$\Delta\theta_{\text{T0}}^{\text{emp}} = 0.0397^\circ \approx 0.04^\circ \quad (18)$$

The experiment measures $(0.04 \pm 0.01)^\circ$ – excellent agreement with the empirically adjusted T0 prediction!

Physical Interpretation of the T0 Correction

The additional rotation arises from three coupled effects:

1. Local Time Field Variation: The intrinsic time field $T(x, t)$ varies along the moving object:

$$T(\vec{r}, t) = T_0 \exp \left(-\xi \frac{|\vec{r} - \vec{v}t|}{ct_H} \right) \quad (19)$$

where $t_H = 1/H_0$ is the Hubble time.

2. Mass-Time Coupling: Through the duality $T \cdot E = 1$, time field variation leads to mass variation:

$$\frac{\delta m}{m} = -\frac{\delta T}{T} = \xi \frac{|\vec{r} - \vec{v}t|}{ct_H} \quad (20)$$

3. Light Deflection by Mass Gradient: The mass gradient acts like a variable refractive index:

$$\frac{d\theta}{ds} = \frac{1}{c} \nabla_\perp \left(\frac{GM_{\text{eff}}(s)}{r} \right) = \xi \frac{1}{c} \nabla_\perp \left(\frac{\delta m}{m} \right) \quad (21)$$

Integration over the light path yields the observed additional rotation.

Connections to Other Phenomena

The T0-modified Terrell-Penrose effect has implications for:

High-Energy Astrophysics: Relativistic jets from AGN should show:

$$\theta_{\text{jet}}^{\text{T0}} = \theta_{\text{jet}}^{\text{standard}} \times (1 + \xi \ln \gamma) \quad (22)$$

Particle Accelerators: In collisions with $\gamma > 1000$ (LHC):

$$\Delta\theta_{\text{LHC}} \approx \xi \times 90^\circ \times \ln(1000) \approx 0.09^\circ \quad (23)$$

Cosmological Distances: Galaxies at $z \sim 1$ should show apparent rotation of:

$$\theta_{\text{gal}} = \xi \times 180^\circ \times \ln(1 + z) \approx 0.05^\circ \quad (24)$$

measurable with JWST/ELT.

.4 Cosmology Without Expansion

T0 postulates NO cosmic expansion, similar to Steady-State models [37, 38] and modern alternatives [41, 40].

Redshift Through Time Field Evolution

Redshift arises through frequency-dependent shifts:

$$z = \xi \ln \left(\frac{T(t_{\text{beob}})}{T(t_{\text{emit}})} \right) \quad (25)$$

This resembles "Tired Light" theories [39], but avoids their problems through coherent time field evolution.

CMB Without Inflation

CMB temperature fluctuations arise from quantum fluctuations in the time field, without inflationary expansion [17]:

$$\frac{\delta T}{T} = \xi \sqrt{\frac{\hbar}{m_{\text{Planck}} c^2}} \approx 10^{-5} \quad (26)$$

This solves the horizon problem without inflation, similar to Variable Speed of Light theories [42, 43].

.5 Experimental Evidence

High-Energy Physics

- LHC Jet Quenching: $R_{AA} = 0.35 \pm 0.02$ with T0 correction [28, 32]
- Top Quark Mass: $m_t = 172.52 \pm 0.33$ GeV [29]
- Higgs Couplings: Precision $< 5\%$ [31]

Cosmological Tests

- Surface Brightness: $\mu \propto (1+z)^{-0.001 \pm 0.3}$ instead of $(1+z)^{-4}$ [40]
- Angular Sizes: Nearly constant at high z [41]
- BAO Scale: $r_d = 147.8$ Mpc without CMB priors [34]

Precision Tests

- Atom Interferometry: $\Delta\phi/\phi \approx 5 \times 10^{-15}$ expected [66]
- Optical Clocks: Relative drift $\sim 10^{-19}$ [67, 68]
- Gravitational Waves: LISA sensitivity to ξ -modulation [69]

.6 Theoretical Connections

T0 has connections to:

- Loop Quantum Gravity [44, 46]
- String Theory/M-Theory [47, 48]
- Emergent Gravity [60, 61]
- Fractal Spacetime [62, 63]
- Information-Theoretic Approaches [64, 65]

.7 Conclusion

Mass variation is the geometric dual of time dilation in T0 — rigorously equivalent and ontologically unified. The geometrically determined parameter $\xi = 4/3 \times 10^{-4}$ determines all natural constants. T0 explains the Terrell-Penrose effect, the muon $g-2$ anomaly, and cosmological observations without expansion. This addresses historical critiques [3, 4] and modern challenges [35, 36].

Bibliography

- [1] Einstein, A. (1905). Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 17, 891.
- [2] Lorentz, H. A. (1904). Electromagnetic phenomena in a system moving with any velocity smaller than that of light. *Proc. Roy. Netherlands Acad. Arts Sci.*, 6, 809.
- [3] Israel, H., Ruckhaber, E., Weinmann, R. (Eds.) (1931). Hundert Autoren gegen Einstein. Leipzig: Voigtländer.
- [4] Dingle, H. (1972). Science at the Crossroads. London: Martin Brian & O’Keeffe.
- [5] Gift, S. J. G. (2010). One-way light speed measurement using the synchronized clocks of the global positioning system (GPS). *Physics Essays*, 23(2), 271-275.
- [6] Terrell, J. (1959). Invisibility of the Lorentz Contraction. *Physical Review*, 116(4), 1041-1045.
- [7] Penrose, R. (1959). The apparent shape of a relativistically moving sphere. *Proc. Cambridge Phil. Soc.*, 55(1), 137-139.
- [8] Hossenfelder, S. (2025). The Terrell-Penrose Effect Finally Caught on Camera [Video]. YouTube. <https://www.youtube.com/watch?v=2IwZB9PdJVw>.
- [9] Terrell, A. et al. (2025). A Snapshot of Relativistic Motion: Visualizing the Terrell-Penrose Effect. *Nature Communications Physics*, 8, 2003.
- [10] Weiskopf, D., et al. (2000). Explanatory and illustrative visualization of special and general relativity. *IEEE Trans. Vis. Comput. Graphics*, 12(4), 522-534.
- [11] Müller, T. (2014). GeoViS—Relativistic ray tracing in four-dimensional spacetimes. *Computer Physics Communications*, 185(8), 2301-2308.
- [12] Pascher, J. (2025a). T0-Theorie der Zeit-Masse-Dualität [Repository]. GitHub. .
- [13] Pascher, J. (2025b). Quantenmechanik in T0-Framework. 034_T0_QM-optimierung_De.pdf.
- [14] Pascher, J. (2025c). Relativitätserweiterungen in T0. 020_T0_QM-QFT-RT_De.pdf.
- [15] Pascher, J. (2025d). SI-Einheiten und T0. 013_T0_SI_De.pdf.
- [16] Pascher, J. (2025e). Myon g-2 in T0. 018_T0_Anomale-g2-10_De.pdf.
- [17] Pascher, J. (2025f). CMB in T0. 039_Zwei-Dipole-CMB_De.pdf.
- [18] Pascher, J. (2025g). Casimir-Effekt in T0. 091_Casimir_De.pdf.
- [19] Pascher, J. (2025h). Kosmologie in T0. 025_T0_Kosmologie_De.pdf.
- [20] Pascher, J. (2025i). Feinstrukturkonstante aus ξ . 011_T0_Feinstruktur_De.pdf.

- [21] Pascher, J. (2025j). Gravitationskonstante aus ξ . 012_T0_Gravitationskonstante_De.pdf.
- [22] Hafele, J. C., & Keating, R. E. (1972). Around-the-World Atomic Clocks. *Science*, 177(4044), 166-168.
- [23] Ashby, N. (2003). Relativity in the Global Positioning System. *Living Rev. Relativity*, 6, 1.
- [24] Rossi, B., & Hall, D. B. (1941). Variation of the Rate of Decay of Mesotrons with Momentum. *Phys. Rev.*, 59(3), 223.
- [25] Particle Data Group. (2024). Review of Particle Physics. *Prog. Theor. Exp. Phys.*, 2024, 083C01.
- [26] Muon g-2 Collaboration. (2023). Measurement of the Positive Muon Anomalous Magnetic Moment to 0.20 ppm. *Phys. Rev. Lett.*, 131, 161802.
- [27] Fermilab Muon g-2 Collaboration. (2023). Final Report. FERMILAB-PUB-23-567-T.
- [28] CMS Collaboration. (2024). Jet quenching in PbPb collisions. *Phys. Rev. C*, 109, 014901.
- [29] CMS Collaboration. (2023). Top quark mass measurement. *Eur. Phys. J. C*, 83, 1124.
- [30] ATLAS Collaboration. (2023). Muon reconstruction and identification. *Eur. Phys. J. C*, 83, 681.
- [31] ATLAS Collaboration. (2023). Higgs boson couplings. *Nature*, 607, 52-59.
- [32] ALICE Collaboration. (2023). Quark-gluon plasma properties. *Nature Physics*, 19, 61-71.
- [33] Planck Collaboration. (2018). Planck 2018 results. VI. *Astron. Astrophys.*, 641, A6.
- [34] DESI Collaboration. (2025). Baryon Acoustic Oscillations DR2. *MNRAS*, submitted.
- [35] Riess, A. G., et al. (2022). Comprehensive Measurement of H_0 . *ApJ Lett.*, 934, L7.
- [36] Di Valentino, E., et al. (2021). In the realm of the Hubble tension. *Class. Quantum Grav.*, 38, 153001.
- [37] Hoyle, F. (1948). A New Model for the Expanding Universe. *MNRAS*, 108, 372.
- [38] Bondi, H., & Gold, T. (1948). The Steady-State Theory. *MNRAS*, 108, 252.
- [39] Zwicky, F. (1929). On the redshift of spectral lines. *PNAS*, 15(10), 773.
- [40] Lerner, E. J. (2014). Surface brightness data contradict expansion. *Astrophys. Space Sci.*, 349, 625.
- [41] López-Corredoira, M. (2010). Angular size test on expansion. *Int. J. Mod. Phys. D*, 19, 245.
- [42] Albrecht, A., & Magueijo, J. (1999). Time varying speed of light. *Phys. Rev. D*, 59, 043516.
- [43] Barrow, J. D. (1999). Cosmologies with varying light speed. *Phys. Rev. D*, 59, 043515.
- [44] Rovelli, C. (2004). Quantum Gravity. Cambridge University Press.
- [45] Thiemann, T. (2007). Modern Canonical Quantum General Relativity. Cambridge University Press.

- [46] Ashtekar, A., & Lewandowski, J. (2004). Background independent quantum gravity. *Class. Quantum Grav.*, 21, R53.
- [47] Polchinski, J. (1998). *String Theory*. Cambridge University Press.
- [48] Becker, K., Becker, M., & Schwarz, J. H. (2007). *String Theory and M-Theory*. Cambridge University Press.
- [49] Mach, E. (1883). *Die Mechanik in ihrer Entwicklung*. Leipzig: Brockhaus.
- [50] Sciama, D. W. (1953). On the origin of inertia. *MNRAS*, 113, 34.
- [51] Wheeler, J. A. (1990). Information, physics, quantum. In: Zurek, W. (Ed.), *Complexity, Entropy, and Physics of Information*.
- [52] Barbour, J. (1999). *The End of Time*. Oxford University Press.
- [53] Penrose, R. (2004). *The Road to Reality*. Jonathan Cape.
- [54] Penrose, R. (1967). Twistor algebra. *J. Math. Phys.*, 8(2), 345.
- [55] Mandelbrot, B. B. (1982). *The Fractal Geometry of Nature*. W. H. Freeman.
- [56] Di Francesco, P., et al. (1997). *Conformal Field Theory*. Springer.
- [57] Weinberg, S. (2008). *Cosmology*. Oxford University Press.
- [58] CODATA. (2019). Fundamental Physical Constants. *Rev. Mod. Phys.*, 93, 025010.
- [59] Newell, D. B., et al. (2018). The CODATA 2017 values. *Metrologia*, 55, L13.
- [60] Verlinde, E. (2011). On the origin of gravity. *JHEP*, 2011, 29.
- [61] Jacobson, T. (1995). Thermodynamics of spacetime. *Phys. Rev. Lett.*, 75, 1260.
- [62] Nottale, L. (1993). *Fractal Space-Time and Microphysics*. World Scientific.
- [63] El Naschie, M. S. (2004). A review of E infinity theory. *Chaos, Solitons & Fractals*, 19(1), 209.
- [64] Susskind, L. (1995). The world as a hologram. *J. Math. Phys.*, 36, 6377.
- [65] Maldacena, J. (1998). The large N limit of superconformal field theories. *Adv. Theor. Math. Phys.*, 2, 231.
- [66] Kasevich, M. A., et al. (2023). Atom interferometry. *Rev. Mod. Phys.*, 95, 035002.
- [67] Ludlow, A. D., et al. (2015). Optical atomic clocks. *Rev. Mod. Phys.*, 87, 637.
- [68] Brewer, S. M., et al. (2019). Al⁺ quantum-logic clock. *Phys. Rev. Lett.*, 123, 033201.
- [69] LISA Consortium. (2017). Laser Interferometer Space Antenna. arXiv:1702.00786.
- [70] Siehe [3].